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Chinese	Native Speaker
English	TOEFL 107
French	TCF B2

Education

2024-on	Engineer Deg., Télécommunications , Télécom Paris, Île-de-France, France. GPA: 3.95/4.0 Core Courses: Large-scale Digital Infrastructure, Data Science, Graph Learning
2022-2025	M.Eng, College of Electronic and Information Engineering , Tongji University, Shanghai, China. GPA: 4.83/5.0 Ranking: 1/48 <i>Summa Cum Laude</i> Research Area: Internet of Things, Wireless Communications
2018-2022	B.Eng, School of Cyberspace , Hangzhou Dianzi University, Zhejiang, China. GPA: 4.72/5.0 Ranking: 1/116 <i>Summa Cum Laude</i> Core Courses: Computer Networks, Principles of Communication, Signals and Systems

Research Experience

2021-on	Resonant Beam System Project member, researcher, and simulation program developer. Investigating intra-cavity laser resonance for wireless power transfer, communication, and positioning. Developing interpretable analytical models and simulation tools based on Fourier optics and electromagnetic wave propagation for system design and performance evaluation. Analyzing wireless channel characteristics, beam propagation, focusing, and transmission performance. Studying the impact of system and channel parameters on link performance through analytical derivation and numerical simulation. Developing computational frameworks for efficient evaluation and optimization of beam transmission and wireless system performance. Investigating system extensions toward mmWave communication and RIS/IRS-assisted wireless systems, with an emphasis on beamforming and wave manipulation. Research Skills: Python, MATLAB, Fourier Optics, Electromagnetic Wave Propagation, Beamforming, Wireless Communication.
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Publication Highlights

2025	X. Wang, S. Xia , W. Fang, M. Liu, M. Xiong and Q. Liu, " Mobile Self-Protection Resonant Beam SWIPT With Adaptive Phase Control ," in <i>IEEE Internet of Things Journal</i> , vol. 12, no. 20, pp. 43103-43118, 15 Oct.15, 2025.
2025	S. Xia , Q. Liu, Q. Jiang, W. Fang and M. Liu, " Frequency Division Duplexing Resonant Beam Communication ," in <i>IEEE Transactions on Wireless Communications</i> , vol. 25, pp. 6145-6159, 2026, doi: 10.1109/TWC.2025.3623890.
2024	S. Xia et al., " Millimeter-Wave Resonant Beam SWIPT ," in <i>IEEE Internet of Things Journal</i> , vol. 11, no. 24, pp. 40464-40477, 15 Dec.15, 2024.
2023	S. Xia , Q. Liu, M. Liu, W. Fang, M. Xiong, Y. Bai, X. Li, " Auto-Protection for Resonant Beam SWIPT in Portable Applications ," in <i>IEEE Internet of Things Journal</i> , vol. 11, no. 3, pp. 4127-4138, 1 Feb.1, 2024.
2022	M. Liu, S. Xia , M. Xiong, M. Xu, Q. Liu and H. Deng, " NLOS Transmission Analysis for Mobile SLIPT Using Resonant Beam ," in <i>IEEE Transactions on Wireless Communications</i> , vol. 23, no. 1, pp. 334-346, Jan. 2024.
2022	M. Liu, S. Xia , M. Xiong, M. Xu, W. Fang, Q. Liu, " Integrated Communication and Positioning With Resonant Beam ," in <i>IEEE Transactions on Wireless Communications</i> , vol. 21, no. 11, pp. 9186-9199, Nov. 2022.

Patents

2023 Q. Liu, S. Xia, M. Liu, Y. Bai, S. Han, M. Xu, X. Li, "**A Portable Auto-Protected Simultaneous Wireless Information and Power Transfer System**," CN116707652A, Sep. 05, 2023. Under examination as to substance.

2023 Q. Liu, X. Li, S. Xia, "**A Barcode Scanning System based on Resonant Beam System**," CN1172773030A, Dec. 22, 2023. Under examination as to substance.

2023 Q. Liu, Y. Zhu, S. Xia, Q. Jiang, S. Shen, "**A Priority-based Dynamic Communication Resource Allocation Method and Apparatus**," CN116347634A, Jun. 27, 2023. Under examination as to substance.

2021 W. Yin, Q. Zheng, H. Hu, S. Xia, K. Ye, X. Zhang, L. Zhai, T. Wu, "**An Intrusion Detection Method and Apparatus based on Modbus Packet Reassembly**," CN114124447A, Mar. 01, 2022. Patent granted in 2024.

2020 Q. Chen, T. Wu, Q. Zheng, L. Dong, C. Hu, S. Xia, et., al., "**A Lightweight Certificate-Free Industrial Internet of Things Access Control Method and System**," CN111526130A, Aug. 11, 2020. Patent granted in 2021.

Work Experience

2026–2026 **Wireless Communication Engineer**, VNEXIA Technology, Paris, France.
Architected a Private Sub-GHz wireless backbone for Edge AI nodes, utilizing a custom API-proxy gateway and PHY-layer optimization to enable real-time cloud synchronization in high-interference "white zones".

2025–on **Peer Reviewer**.
IEEE Journal on Selected Areas in Communications, and *IEEE Internet of Things Journal*.

2023–2025 **Academic Mentor**, Tsinghua University, Beijing, and Chase Education Co., Ltd., Hubei, China.
Mentoring undergraduate students in conducting research projects related to computer science.

2022–2025 **Research Assistant**, Tongji University, Shanghai, China.
Conducting research under China national key technology and innovation grant projects.

2019–2022 **Teaching Assistant**, Hangzhou Dianzi University, Zhejiang, China.
Teaching courses: *Signals and Systems*, *Steganography and Digital Watermarking*.

2021–2022 **President of Student Union**, Hangzhou Dianzi University, China.
Responsible for internal and financial affairs, and the Dept. Cultural & Academic Activities.

Honors

2025 **Outstanding Graduates**. Municipal Education Commission, Shanghai, China.

2024 **France Excellence EIFFEL Scholarship**. Campus France, France.

2024 **National Scholarship for Master Students**. Ministry of Education, China.

2023 **National Scholarship for Master Students**. Ministry of Education, China.

2022 **Valedictorian of the University Graduation Ceremony**. Hangzhou Dianzi University, Zhejiang, China.

2022 **Outstanding Graduates**. Department of Education, Zhejiang, China.

2021 **Zhejiang Provincial Scholarship**. Department of Education, Zhejiang, China.

2021 **Top-10 Students of the Year**. Hangzhou Dianzi University, Zhejiang, China.

2020 **National Scholarship for Undergraduate Students**. Ministry of Education, China.

2019 **Zhejiang Provincial Scholarship**. Department of Education, Zhejiang, China.